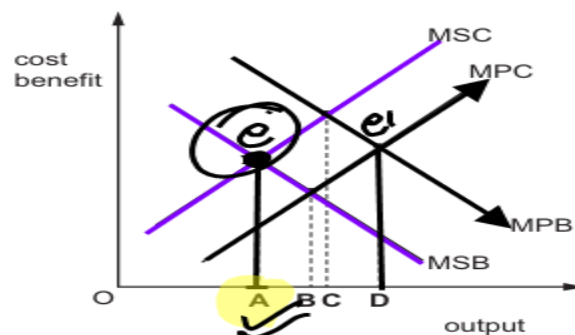


1-A:

Socially optimal equilibrium is where Marginal social cost is equal to Marginal social benefit. This is shown through the equilibrium “e” and the corresponding level of output is “A”. This explains why option A will be correct.

On the other hand, private equilibrium is where $MPC=MPB$. It is shown through the output level “D”



2-C:

A – Producers do not under-value the benefits of a public good; this is because due to the free-rider problem, they do not even involve themselves in providing these goods. Instead, under-realization by producers is generally seen in the case of “merit goods”

B – Even if public goods have high fixed costs, this is not a cause of market failure.

C – Public goods are non-excludable, that is when, payment by one individual does not restrict others from enjoying the benefits associated with these goods; even the non-tax payers get to consume them, which results in a “free-rider problem”, that is when consumers will never be willing to pay for these goods because they are aware that they can enjoy the equal benefits of these goods, without paying for it, that tax-payers get to enjoy.

As a result of which, private producers will never provide for it, and this non-provision of public goods results in market failure.

D – Public goods are non-rivalrous in consumption, that is when, consumption by one individual will not reduce the amount available for others to consume.

3-C:

- ➔ The cost of building the railway line will be borne by the party , who is directly involved in producing it, which in this case is the government. Therefore, this will be classed as a “private cost”
- ➔ An increase in noise levels represents an external “cost” , that is the cost to the third party, for which no appropriate compensation is to be made.
- ➔ This explains, why key C, will be correct.

4-B:

A – Marginal utility cannot be kept constant when deriving an individual demand curve. This is because the demand curve is the same as marginal utility curve.

B – The “consumer’s income” is an “other” factor that is assumed to remain constant when deriving an individual’s demand curve.

C – The price of the product and the quantity demanded can vary. Whereas, all other factors that affect demand except for the price factor are assumed to remain unchanged.

D – Total utility is derived through the marginal utility curve, which is the same as the demand curve. This is because price affects quantity demanded, which in turn affects “marginal utility”. And since, marginal utility changes, total utility will also change.

5-B:

At point “B”, since “total utility is at its maximum”, it can be said that marginal utility is **equal to “zero”**. This is why, this becomes correct.

6-D:

By imposing a sales tax of 25% on both the goods, the prices of both the goods must increase because of which, the quantity demanded of both the goods must fall.

A – Here the quantities of both the goods are rising which represents that there must have been a rise in real-income. This may have occurred because of a fall in the price of both goods or because of an increase in nominal income.

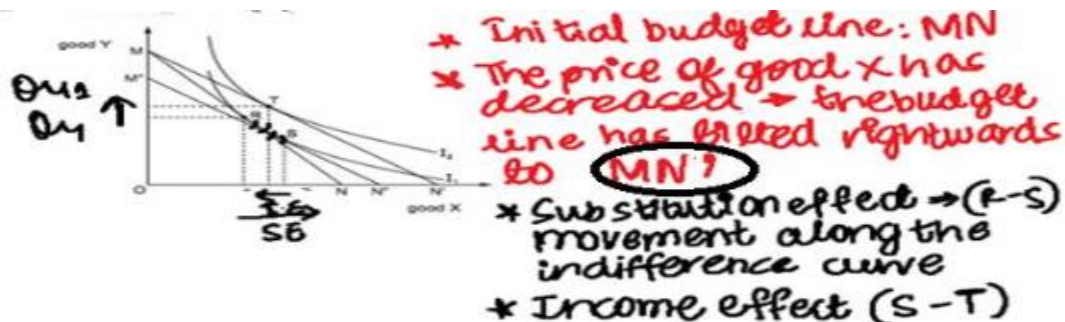
B – Here the quantity of good X is shown to rise which represents that the price of good X is falling.

C – Here the quantity of good Y is shown to rise which represents that the price of good Y is falling.

D – Here the quantities of both the goods is shown to fall, which represents that the price of both the goods may have increased. This will be a result of imposition of sales tax of 25% on both the goods.

This is why key D is correct.

7-A:



Since the income effect and the substitution effect move in the opposite direction, it can be understood that good X is an inferior good.

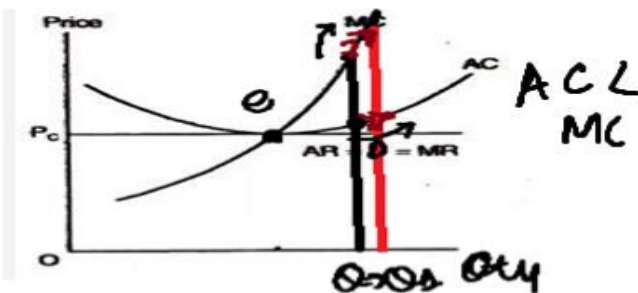
Since the price of good X has fallen, and as a result of which, quantity demanded of good Y has risen, it can be concluded that X and Y are complements, because there is a negative relationship between the price of one good and the qty of another in case of complements.

This is why, key A, is correct.

8-D:

-> It can be seen in the diagram below that a firm producing a level of output where average cost is lower than the marginal cost, when increases its output by one unit, will face a rise in both: average cost and marginal cost.

This is why, key D, becomes correct.



9-B:

A) This option is unlikely to affect small local bread producers significantly. As they typically don't rely heavily on television advertising, an increase in costs would not directly impact their business. **Instead, large producers would be affected from it.**

- B) This is a significant threat, as small local bread producers often focus on quality and uniqueness. If consumers prioritize price over these factors, they may switch to cheaper mass-produced bread, reducing demand for local products and threatening the survival of small bread producers. This explains why key B is correct.

C) This option is more likely to affect large industrial bakeries than small local bread producers. Diseconomies of scale occur when a business grows too large, leading to inefficiencies that translate into an increase in long-run average costs.

D) Many small local bread producers likely deliver to local customers or sell through their own shops, minimizing the impact of road congestion. Importantly, this is an example of an external diseconomies of scale.

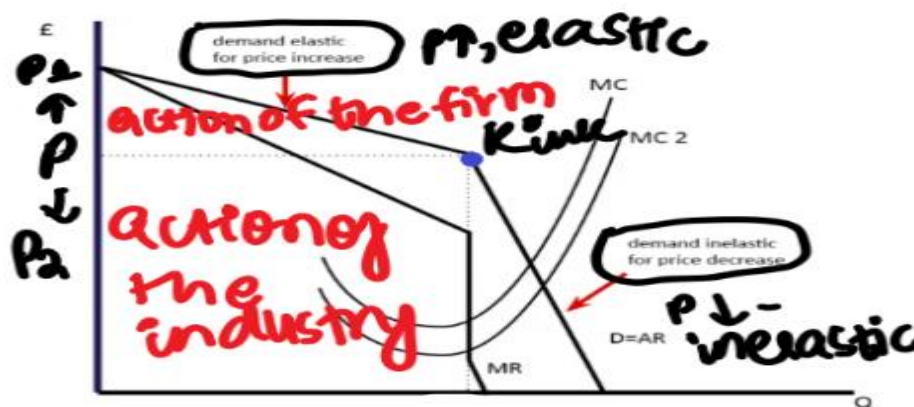
10-B:

- Firms advertise their products with **different brand names**, indicating **product differentiation**, which is a characteristic of **monopolistic competition**.

- Some firms make more profit than others in the short run, suggesting that there are temporary profits (**super normal profits**) to be made, but in the long run, all firms make normal profit, indicating that the market is competitive and profits are driven back to normal levels.
- Perfect competition (A) is characterized by no product differentiation and no advertising, so it doesn't fit.
- Oligopoly (C) is characterized by a small number of large firms, who make supernormal profits in the short-run and the long-run.
- Monopoly (D) is a single-firm market, so it doesn't fit either.

In **monopolistic competition**, firms have some market power due to product differentiation, but there are still many firms in the market, and entry is relatively easy, leading to normal profits in the long run.

11-A:



If for example, the firm raises its price, no other firm in the industry will follow, and therefore, this represents “**elastic demand**” (because, it is the behavior of a firm), and as a result, there is a contraction in quantity demanded by a relatively higher percentage, which results in a “fall in Total Revenue”. **This is why, key A, is correct**, because it explains **why a firm would not want to increase its price above “P”**

Importantly, when a firm decreases its price below “P”, all other firms in the industry will follow, and therefore, this represents “**inelastic demand**” (because, it is the behavior of an industry), and as a result, there will be an extension in qty demanded by a relatively smaller percentage, which results in a “fall in Total Revenue”

→ **Therefore, prices remain rigid or unchanged at the “kink”**

This explains why key A is correct.

12- C:

Firm’s total revenue will be maximized **when marginal revenue is zero**. This is illustrated through the output level C, which explains why key C is correct.

A – Here Marginal revenue is equal to Average cost, which explains why this is incorrect.

B – Here Marginal revenue is equal to Marginal cost, which represents the profit maximizing level of output.

D – Here, Average revenue is equal to zero.

13-A:

A) A limit on the amount by which energy prices can rise represents a “maximum price legislation” in the case of which, there will be a further extension in qty demanded of energy. And when, quantity demanded of energy extends relative to quantity supplied of energy which contracts, there will be a shortage of energy, which will further worsen the issue. This is why, this becomes correct.

B) An increase in the tax on energy usage will help to internalize the negative externalities associated with excessive energy usage, and when this results in a reduction in qty traded of energy to the socially optimum level of output, this will represent an efficient usage of energy.

C) Energy pricing policies that help to reflect the full social cost of energy means that both: private costs and external costs, will be considered when setting prices. This will also help to internalize the negative externalities associated with excessive energy usage, and when this results in a reduction in qty traded of energy to the socially optimum level of output, this will represent an efficient usage of energy.

D) When subsidies are provided to suppliers of products that are designed to save energy, as a result of which, the prices of these energy-saving substitutes decrease, this will lead to consumers to switch to them, encouraging a fall in the consumption of products that result in more energy usage. As a result of which, an efficient usage of energy will be achieved.

14-C:

Intergenerational equity, also known as **intergenerational justice or fairness**, refers to the principle of fairness and justice between different age groups or generations. It aims to ensure that resources, benefits, and opportunities are distributed fairly and sustainably across generations, without unfairly burdening or benefiting one generation at the expense of another.

- The proposal aims to reduce income inequality between generations by redistributing resources from wealthy elderly individuals (who have high levels of real income) to young adults in need. This redistribution of resources from one generation to another will increase intergenerational equality. **This redistribution would increase intergenerational equity, as it addresses income disparities between age groups.**

15-A:

A) Economic rent is the payment made to a factor of production that is in excess of the minimum amount necessary to keep the factor in its present occupation. This definition matches the question, which describes a payment "over and above that necessary to keep the factor in its present use". Therefore, option A, is the correct answer.

B) Normal profit represents a situation where zero-economic profits are made that occurs when Average revenue is equal to Average cost.

C) Opportunity cost is the value of the next best alternative that is given up when a choice is made. It is not the payment made to a factor of production over and above the minimum necessary.

D). Transfer earnings are the minimum amount necessary to keep a factor of production in its present use, not the payment made over and above that amount. Therefore, D) Transfer earnings is not the correct answer.

16-C:

The correct answer is C) It increases at a falling rate.

When the marginal product of labor decreases but remains positive, it means that each additional worker is contributing to an increase in total output, but the amount of the increase is getting smaller. This is a classic case of **diminishing marginal returns**.

As a result, the total output of the firm will continue to increase, but at a decreasing rate. **This means that the firm's total output will increase, but the rate of increase will slow down as more workers are added. (In other words, total product increases at an decreasing rate)**

A) It stops increasing: This would happen if the marginal product of labor became zero which represents that total product has maximized and cannot rise further.

B) It increases at a constant rate: This would happen if the marginal product of labor remained constant, which is not the case in this scenario.

D) It increases at a rising rate: This would happen if the marginal product of labor increased as more workers were added, which is not the case in this scenario.

17-C:

Note that a leftward shift of the supply curve of labor means that the quantity of labor supplied decreases at each wage level, while a downward movement along the supply curve means that the quantity of labor supplied decreases in response to a decrease in the wage.

C: A leftward shift of the supply curve of labor in a particular industry means that at any given wage, fewer workers are willing to supply their labor to that industry. This can happen if wages in other industries that offer similar jobs increase, making those industries more attractive to workers and causing them to leave the original industry.

A) A fall in demand for the industry's product: This would lead to a decrease in the demand for labor since demand for labor is derived demand, but it would not shift the supply curve of labor.

B) A greater use of machinery in the production of the industry's output: This could lead to a decrease in the demand for labor, but it would not shift the supply curve of labor.

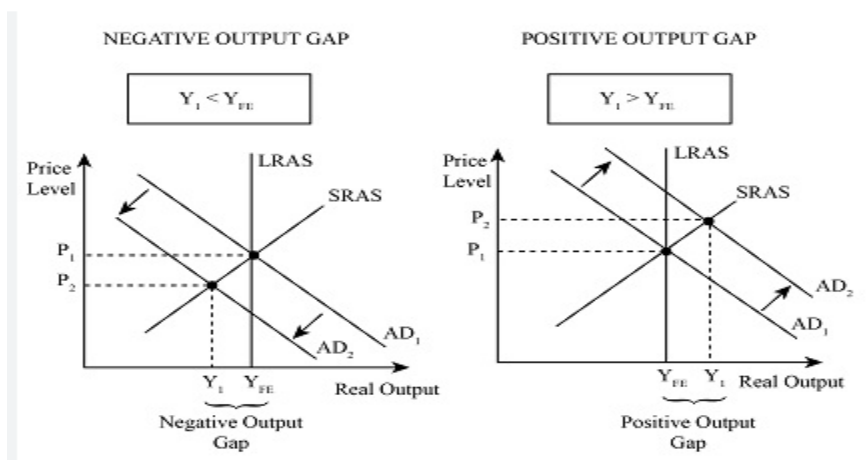
D) A weakening of trade union influence in the industry: This could lead to a decrease in wages, but it would not shift the supply curve of labor.

18-A: The output gap is a measure of the difference between the actual output of an economy and its potential output. Potential output is the maximum amount of goods and services an economy can produce when it is at full capacity. An output gap suggests that an economy is running at an inefficient rate, either producing more than it can sustain in the long term (**positive output gap**) or producing less than it is capable of producing (**negative output gap**). **This is why, key A, is correct.**

B) "Output when labor is fully employed" is not the same as "long-run potential output". All other resources apart from labor also need to be fully employed too to achieve that.

C) This option is describing the difference between the highest and lowest points of the business cycle, not the output gap.

D) This option is also describing the business cycle. It is not describing the output gap.



19-C:

The correct answer is C) The size of the population may have changed over the period.

GNP (Gross National Product) per head is a measure of the total output of a country divided by its population. Since the data is in “per-head”, this represents that the size of the population is already accounted for, which makes this the incorrect option.

A) The increase in GNP per head may have been concentrated among a small group of people, while the majority of the population may not have seen an increase in their living standards.

B) If prices have increased, the purchasing power of **the increased GNP per head** may be reduced, meaning that **average living standards may not have risen as much as the increase in GNP per head suggests**.

D) If people are working longer hours to achieve the increase in GNP per head, it may not necessarily translate to an increase in average living standards, as the benefits of the increased output may be offset by the decreased leisure time.

20-D:

Human development index includes:

GNI per capita

Education: mean years of schooling and average years of schooling

Health: Life expectancy at birth.

It does not include: child mortality per thousand population, which is why, key D becomes correct.

21-A:

- A) A decrease in labor costs per unit of output in Asian countries would make their exports cheaper and more competitive in the US market, leading to an increase in imports and a decline in demand for domestically produced manufactured goods. This would lead to structural unemployment in the US manufacturing sector, as US firms would struggle to compete with the cheaper imports. **This represents “international unemployment”**
- B) This would make imports more expensive and less competitive, potentially leading to an increase in demand for domestically produced goods and a decrease in structural unemployment mainly **“international unemployment”**
- C) This could lead to a decrease in wages of the employees in the US which will translate into a fall in costs of production, as a result of which, the goods in the US would appear competitive, preventing demand from switching to imports.
- D) A weaker US dollar would make exports cheaper and more competitive, potentially leading to an increase in demand for US manufactured goods and an increase in export revenues lead to a rise in Aggregate demand which subsequently translates into a decrease in demand deficient unemployment.

22-D:

Note: Frictional unemployment refers to the temporary unemployment that occurs when workers are transitioning between jobs. Supply-side policies aim to reduce this type of unemployment by making the labor market more efficient.

D - Supply-side policies improve the efficiency of labor through increased investment on education and training which is an interventionist supply side tool. Since this makes workers more flexible and occupationally mobile, they can easily switch in between occupations when the structure of the economy is changing, as a result of which, structural unemployment that is classed under natural rate of unemployment decreases.

A) Governments can implement policies to reduce seasonal unemployment, such as providing training programs or incentives for businesses to hire workers during off-peak seasons.

B) Changing interest rates can affect demand-deficient unemployment, but it is not an effective way to reduce the natural rate of unemployment, which is influenced by structural factors such as labor market rigidities and skill mismatches.

C): A depreciation of the exchange rate can actually boost export revenues and increase aggregate demand, which can help reduce demand-deficient unemployment.

23-D:

Injects into the circular flow of income are expenditures that enter the economy, increasing aggregate demand and boosting economic activity. The options A, B, and C are all injections:

- A rise in exports of financial services (A) generates export revenue from abroad.
- Firms investing in new production techniques (B) increases investment expenditure.
- Increased government spending on health services (C) is a form of government expenditure.

In contrast, spending on hotels and entertainment by tourists visiting another country (D) is a leakage from the circular flow, as it represents import expenditure flowing out of the economy to another country. This is an import of services, not an injection.

24-A:

To find the real economic growth rate, you need to subtract the inflation rate from the nominal GDP growth rate. This is because nominal GDP growth includes the effects of inflation, while real economic growth represents the actual increase in the production of goods and services.

W: $2.1 \text{ (nominal gdp)} - 1.9 \text{ (inflation)} = 0.2 \text{ (real)}$

X: $-1.6 \text{ (nominal gdp)} - 8.9 \text{ (inflation)} = -10.5 \text{ (real)}$

Y: $4.1 \text{ (nominal gdp)} - 5.8 \text{ (inflation)} = -1.7 \text{ (real)}$

Z: $-2.0 \text{ (nominal gdp)} - 16.4 \text{ (inflation)} = -18.4 \text{ (real)}$

Country W has the highest real economic growth rate, 0.2%

B) X and Z did not experience deflation, as their inflation rates are positive.

C) Y did not experience the greatest rise in the standard of living, as its real economic growth rate is negative.

D) There is no information in the table about public debt, so it's impossible to conclude which country experienced the greatest growth in public debt.

25-A:

Injections include **government spending, investment spending and export revenues**.

$A - I > S$: yes-When investment spending that is an injection into the circular flow is greater than savings that is a withdrawal from the circular flow, this represents “net injections” into the circular flow.

$X < M$: No- When export revenue that is an injection into the circular flow is lesser than import spending which represents a withdrawal, this represents “net leakages” from the circular flow.

$G < T$: No - When government spending that is an injection into the circular flow is lesser than tax revenues which represent a withdrawal, this represents “net leakages” from the circular flow.

26-C:

Since the economy is a closed economy without a government, this represents that a two sector multiplier will be applicable:

Multiplier:- $1/(1-mpc)$ or $1/mps$

26 In a closed economy with no government there is an equilibrium level of national income of \$1000 billion. The households always spend 80% of the income they receive. $\rightarrow MPC = 0.8$

If full employment occurs when national income is \$1200 billion, by how much must aggregate demand initially change to close the economy's deflationary gap?

A -\$200 billion B -\$40 billion C +\$40 billion D +\$200 billion

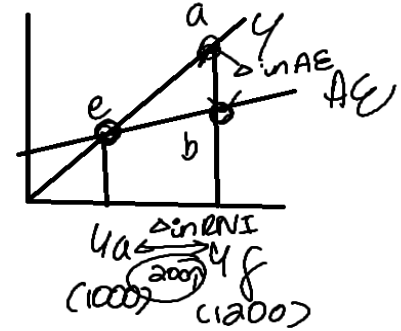
$$\text{Multiplier} = \frac{1}{1-0.8} \rightarrow \frac{1}{0.2} \rightarrow 5$$

$$\text{Increase in AE} \times \text{Multiplier} = \text{Increase in RNI}$$

$$x \times 5 = 200$$

$$5x = 200$$

$$x = \frac{200}{5} \rightarrow 40 \text{ bn } \$$$



27- C:

According to Keynesians, the speculative motive for holding money is the driving force behind the negative correlation between the demand for money and the interest rate.

When interest rates are high, the opportunity cost of holding money is high, and people are less likely to hold liquid cash. Conversely, when interest rates are low, the opportunity cost of holding money is low, and people are more likely to hold liquid cash.

This is also because in the case of speculative motive, money demand is interest elastic.

However, the precautionary and transactionary motives are weakly related to interest rates or in other words, the demand for these motives is "interest inelastic".

28- A:

Macro-economic equilibrium will be where Aggregate demand is equal to Aggregate supply.

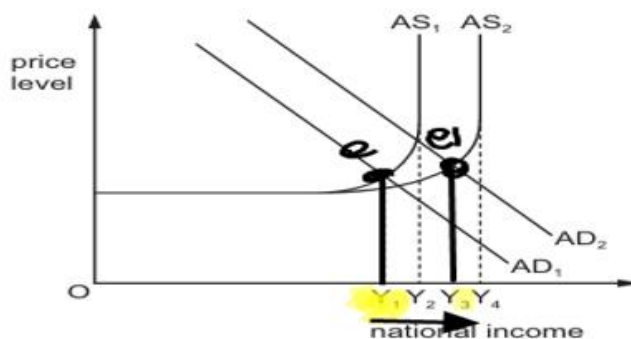
Before the changes, the equilibrium was **e** where AD1 was equal to AS1, and so the equilibrium level of national income was **Y1**.

After the changes, the equilibrium will be "e1" where AD2 was equal to AS2, and so the equilibrium level of national income was Y3.

This implies that the equilibrium level of national income increased from Y1 to Y3.

This can be seen in the diagram below.

There is a change from AD_1 to AD_2 and from AS_1 to AS_2 .



29-B:

The **Phillips Curve** suggests that there is a trade-off, or in other words, a negative relationship between unemployment and inflation, specifically that as unemployment falls, inflation rises.

The correct answer is B) 2011-2012.

The Phillips Curve suggests that as unemployment falls, inflation rises. Between 2011 and 2012, the unemployment rate fell from 6.2% to 5.8%, and the inflation rate rose from 1.6% to 1.8%. This is the only consecutive year pair that shows a decrease in unemployment accompanied by an increase in inflation, which is in agreement with the Phillips Curve analysis.

A) 2010-2011: Unemployment rate falls, but inflation rate also falls (instead of rising, as predicted by the Phillips Curve).

C) 2012-2013: Unemployment rate rises, but inflation rate also rises.

D) 2013-2014: Unemployment rate falls, but inflation remains stable.

The table gives an economy's unemployment rate and inflation rate for a five-year period.

year	unemployment rate %	inflation rate %
2010	7.6	2.6
2011	6.2	1.6
2012	5.8	1.8
2013	5.9	2.0
2014	5.8	2.0

30-D:

To boost economic growth and reduce unemployment, a government would likely adopt expansionary policies, such as:

- Reducing income taxes to increase disposable income and encourage consumption
- Cutting interest rates to lower borrowing costs and stimulate investment

This will subsequently boost Aggregate demand, thereby helping the economy to recover from demand deficient unemployment, which is why, key D will be correct.

A) An appreciation of the currency and higher interest rates would likely worsen deflation and unemployment since these would have a contractionary effect on Aggregate demand.

B) Although an increase in government spending on education will boost Aggregate demand, this will be offset because of an increase in income tax which will decrease disposable income and consumption thereby dampening Aggregate demand.

C) Although a depreciation of the exchange rate will result in a BOT surplus in the long-run when the Marshall Lerner condition is held, which will boost Aggregate demand, an accompanying increase in indirect taxes after having increased prices will bring about a decrease in real-incomes which will dampen consumption expenditure, subsequently decreasing Aggregate demand.

